

FOR OFFICIAL USE



National  
Qualifications  
2018

Mark

**X735/77/01**

**Graphic Communication**

THURSDAY, 10 MAY

1:00 PM – 3:00 PM



\* X 7 3 5 7 7 0 1 \*

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day

Month

Year

Scottish candidate number

**Total marks — 80**

Attempt ALL questions.

All dimensions are in mm.

All technical sketches and drawings use third angle projection.

You may use rulers, compasses or trammels for measuring.

In all questions you may use sketches and annotations to support your answer if you wish.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.



\* X 7 3 5 7 7 0 1 0 1 \*

1. A cafe chain called 'Taco Health' is opening a new premises. A promotional event is being planned prior to the opening and a marketing company has been asked to produce a promotional video that features animated characters.

The purpose of the video is to communicate the following information:

- the main features of the interior design
- how customers use the cafe
- how employees prepare orders and serve the customers.

- (a) Describe two advantages of using each of these techniques in creating the promotional video.

(i) Motion-capture

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(ii) Motion-tweening

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(iii) Transitions

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1. (continued)

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- (b) State a suitable file type for the video to be viewed on a mobile device.

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\_\_\_\_\_  
Samples of printed items used in the cafe will also be available to view at the promotional event. All of these items feature the company logo which includes both graphics and text. The printed items include the following:

- Menus printed on biodegradable plastic
- Window graphics with a height of 2500 mm
- Promotional posters that include a copper spot colour.

- (c) Describe three issues the print technician must address to ensure quality and consistency of the brand across these different printed items.

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The promotional event will also highlight the importance of environmental considerations in the cafe's business model.

- (d) Describe how the cafe's menu would be produced to minimise the impact on the environment, other than by using a biodegradable substrate.

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1. (continued)

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A phone app will also be launched at the event. It will include the location of the new cafe, an online booking facility that includes a floor plan of the new store and a menu tool.

Refer to the **supplementary sheet for use with question 1 (e)** for images of the phone app.

Design elements and principles have been used in the design of the store location mapping tool shown in **Image 1**.

- (e) (i) State the single design principle used to suggest the blue circle in **Image 1** is 'flashing'.

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- (ii) Describe how the flashing blue circle enhances the user's experience of the mapping tool.

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1. (continued)

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A .DXF file format is used to display the floor plan of the new premises in Image 2.

- (f) Explain, giving two reasons, why this file format was chosen.

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Raster file types were used in the app in both the mapping tool in Image 1 and the menu tool in Image 3.

- (g) (i) Describe an advantage of using this file type for the mapping tool.

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- (ii) Describe an advantage of using this file type in the menu tool. You must give a different advantage to the one you have described above.

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[Turn over



2. A chemical engineering company called CENCHEM ENGINEERING INTERNATIONAL has several smaller companies within its corporate group, each with their own logo. Three examples of these are shown below.

Logo 1



Logo 2



Logo 3



- (a) Explain, giving two reasons, why it is desirable for these companies to have logos with simple clear shapes and solid colour fills.

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There are similarities in the design of each of these logos.

- (b) Describe how this has been achieved by making reference to specific design elements or principles, other than those relating to shape or colour.

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2. (continued)

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Each logo includes the registered trade mark symbol shown. ®

- (c) Explain why it is necessary for companies to protect their logo designs in this way.

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CENCHEM ENGINEERING INTERNATIONAL has developed many forms of graphic communications which include both digital and printed media. The main company logo, shown below, appears in all communications.



- (d) Describe four issues associated with replicating this logo across printed and digital media.

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2. (continued)

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Two examples of CENCHEM ENGINEERING INTERNATIONAL company graphics have been included on the supplementary sheet.

Refer to the **supplementary sheet for use with question 2 (e), Image 1 and Image 2.**

**Image 1** An interactive advertisement from the company website aimed at recruiting new graduates.

**Image 2** A printed leaflet aimed at communicating statistical information to the general public.

- (e) (i) Describe, with reference to **Image 1**, how the following design elements and principles have contributed to the creation of a clear focal point in this design.

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Depth of field \_\_\_\_\_

\_\_\_\_\_

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Shape \_\_\_\_\_

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Radial balance \_\_\_\_\_

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- (ii) Explain why the use of a 'focal point' helps the advertisement communicate effectively with this target audience.

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2. (e) (continued)

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- (iii) Describe, with reference to **Image 2**, how the designer has used 'white space', 'grid structure' and 'unity' in the leaflet to communicate effectively with the target audience.

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White space \_\_\_\_\_

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Grid structure \_\_\_\_\_

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Unity \_\_\_\_\_

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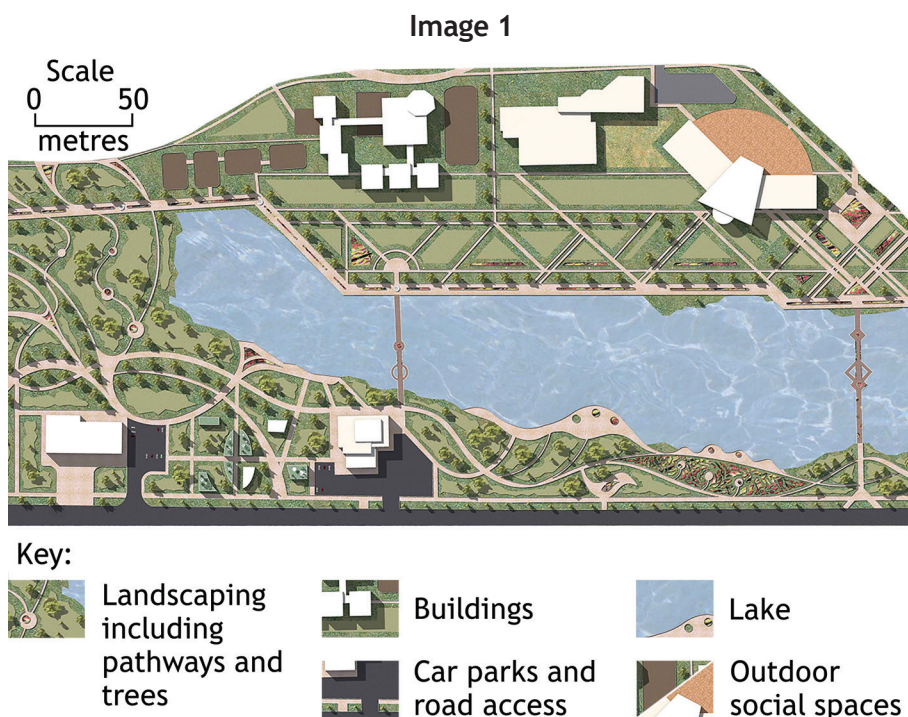
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3. An architectural company has produced plans for a new office and leisure complex. The leisure complex will be available to office staff and local residents. In order to communicate effectively with this target audience an artist's impression of the site plan has been produced, shown in **Image 1**.



- (a) Explain why the artist's impression has been used to communicate the proposal to this target audience.

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A British Standards site plan of the same area is also required to help professionals working on the project.

- (b) (i) State two features that would be included in a British Standards site plan that do not appear in the artist's impression.

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Feature 1 \_\_\_\_\_

Feature 2 \_\_\_\_\_



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3. (b) (continued)

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- (ii) Describe how a landscape architect would make use of the British Standards site plan during the project.

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An underground survey was carried out on the land before construction started. The results of this survey can be useful to various professionals in the building sector.

- (c) (i) State two purposes of this type of survey.

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- (ii) Describe how a structural engineer would make use of an underground survey.

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- (iii) Describe how a conservation body would make use of this type of survey.

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(d) Explain the effects that this discovery would have on the planning chart and on the resource management of the project.

[illegible]

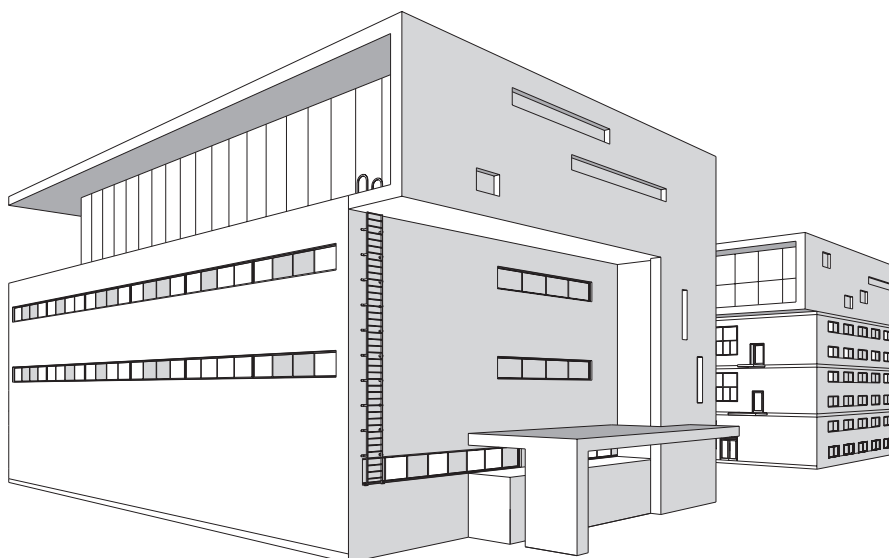
[Turn over for next question

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\* X 7 3 5 7 7 0 1 1 3 \*

4. A 3D computer model of a proposed office unit is shown below. It is to be used in a promotional document where it will be placed in a suitable environment.



- (a) Name four different computer-aided illustration techniques and describe how they would enhance the model of the office unit after it has been placed in a suitable environment.

8

Technique 1 \_\_\_\_\_

Description \_\_\_\_\_

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\_\_\_\_\_

Technique 2 \_\_\_\_\_

Description \_\_\_\_\_

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Technique 3 \_\_\_\_\_

Description \_\_\_\_\_

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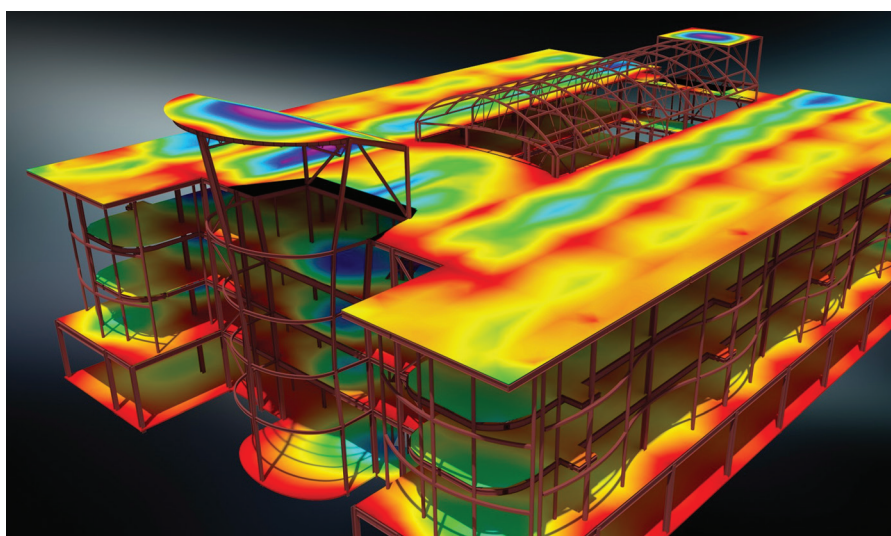
4. (a) (continued)

Technique 4 \_\_\_\_\_

Description \_\_\_\_\_

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A structural engineer has been asked to test certain aspects of the design.  
A screen shot taken during the digital test is shown below.



(b) (i) State the digital testing method shown.

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(ii) Describe how the test results could be used by the design and construction company.

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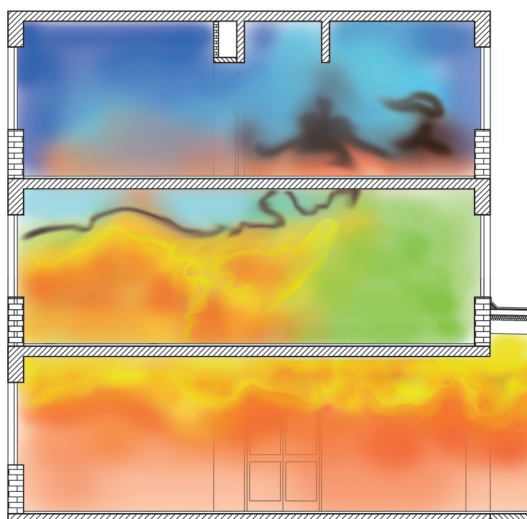
4. (continued)

Digital testing was also requested to simulate the spread of fire through the building.

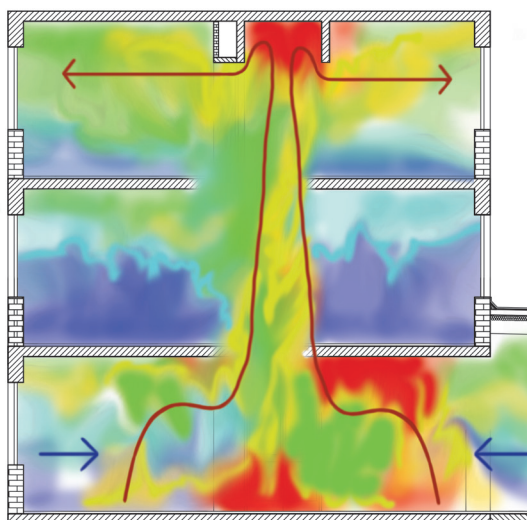
Two screenshots, showing a sectional end elevation of the building, were taken during the testing and simulation. These are shown in **Image 1** and **Image 2** below.

The building has not been drawn to British Standards.

**Image 1: Building  
15 minutes after  
ignition.**



**Image 2: Building  
45 minutes after  
ignition.**





4. (continued)

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- (c) (i) Explain why CFD is the appropriate simulation method to show the spread of fire.

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- (ii) Describe how the data generated by the CFD test could be used, other than in planning an evacuation.

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A fire evacuation route was determined after the company analysed the data from the digital test. A digital simulation was created of the evacuation taking place.

- (d) Explain why the VRML file format was used to save the simulation of the evacuation.

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5. A coffee company is introducing a cafetiere service within its shops. The proposed design of the cafetiere is shown below in **Image 1**.

**Image 1**



The frame of the cafetiere, shown below, in **Image 2** consists of four vertical arms joined together at the base and by a ring at the top. Orthographic views of the frame are shown on the **supplementary sheet** for use with question 5 (a).

**Image 2**

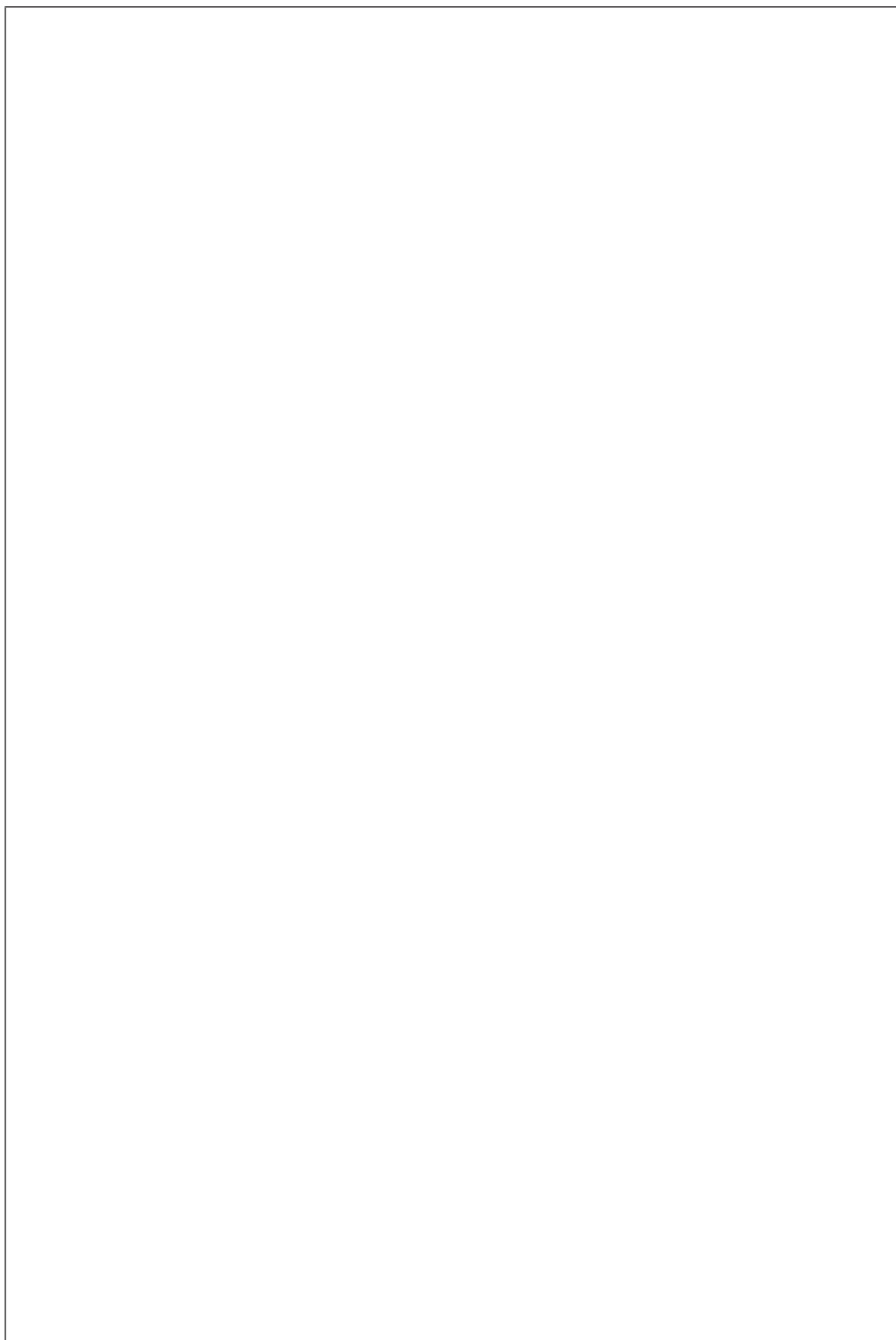


5. (continued)

Refer to the **supplementary sheet** for use with question 5 (a).

- (a) Describe the 3D CAD modelling techniques that would be used to create the frame. You may use sketches to support your answer. Dimensions do **not** need to be included in your response. **Ignore the metal ring and connection to the handle** (these are greyed out on the supplementary sheet).

6




5. (a) (continued)

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5. (continued)

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A model maker has been asked to produce a 3D printed prototype of the handle. This prototype will be used to test the comfort of the handle and how it is assembled to the cafetiere's frame.



- (b) (i) State the name of the file type used to produce a 3D print of the handle. 1

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- (ii) Explain why 'model manipulation' and 'dimensional tolerances' must be considered before 3D printing the handle. 2

Model manipulation \_\_\_\_\_

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Dimensional tolerances \_\_\_\_\_

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[Turn over for next question]



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5. (continued)

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Part of the cafetiere known as the filter spring is shown below. It is designed to keep the filter in touch with the inside of the glass jug.

Refer to the **supplementary sheet** for use with question 5 (c).



- (c) Describe the 3D CAD modelling techniques that would be used to create the filter spring. You may use sketches to support your answer.

3

[END OF QUESTION PAPER]



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ADDITIONAL SPACE FOR ANSWERS AND ROUGH WORK



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## ACKNOWLEDGEMENTS

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